ERIC XU

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EDUCATION

University of California, Santa Cruz — Santa Cruz, CA
M.S. in Applied Mathematics — Incoming — Fall 2026
B.S. in Applied Mathematics — Highest Honors in the Major — June 2026
GPA 3.77 / 4.00 · Dean's Honors (4 quarters) · Graduate coursework as an undergraduate: AM 224, AM 234, AM 261, STAT 266A

TEACHING EXPERIENCE

Undergraduate Tutor & Reader — Baskin School of Engineering, UC Santa Cruz — Fall 2025 – Spring 2026

AM 10 · Mathematical Methods I (Linear Algebra) — Tutor & Reader — 3 quarters

- Led weekly tutoring and office hours for an introductory linear algebra course (matrix operations, vector spaces, eigenvalues/eigenvectors, and linear transformations), turning abstract proofs into worked examples to build student fluency and understanding.
- Graded problem sets and exams as course Reader against instructor rubrics, giving detailed written feedback.
- Rehired for three consecutive quarters across the academic year based on instructor and student feedback.

CSE 16 · Applied Discrete Mathematics — Reader — Winter 2026

- Graded assignments and exams on logic, proof techniques, and combinatorics for a lower-division computer science course.

MATH 19B · Calculus II — Tutor — Winter 2026

- Tutored students in integration techniques, sequences and series, and applications of the definite integral.

Teaching preparation: AM 178A – Learning Assistant Training (completed) · AM 200 – Applied Mathematics TA Training (planned).

RESEARCH & PROJECTS

Undergraduate Researcher — TACS Lab (Prof. Ashesh Chattopadhyay), UC Santa Cruz — September 2025 – Present

- Developing an ML pipeline for "grey swan" tropical cyclone prediction, coupling a GraphCast global weather model with super-resolution downscaling to HRRR (~3 km) resolution.
- Run large-scale training and inference on NCAR (Derecho) and NERSC (Perlmutter) HPC clusters via SLURM and PBS schedulers.

Selected projects:

- **4D-Var Data Assimilation with a Learned Inverse Observation Operator** — senior capstone (AM 170B); final report graded 95/A on Lorenz-96 and Kolmogorov-flow systems.
- **Cross-species transfer learning with the DeepSEA genomic model** (human → mouse ENCODE data), including zero-shot evaluation and baseline pipelines.

TECHNICAL SKILLS

Programming: Python, C/CUDA, MATLAB, R, Bash
Machine Learning & Scientific Computing: PyTorch, JAX, NumPy, Triton (GPU kernels)
HPC & Tools: SLURM, PBS, Linux, Git, SSH
Mathematical Areas: Linear algebra, differential equations, PDEs, dynamical systems, numerical methods, deep learning

RELEVANT COURSEWORK

Applied Mathematics: Linear Algebra (AM 10) · Mathematical Methods II–III (AM 20, 100) · Calculus III (AM 30) · Intro to PDEs (AM 112) · Dynamical Systems (AM 114, 224) · Mathematical Modeling I–II (AM 170A/B) · Scientific Computing (AM 129) · Computational Methods (AM 147) · GPU Programming (AM 148) · Scientific Machine Learning (AM 261) · ML & AI in Genomics (AM 234)

Statistics & Computer Science: Probability & Statistics (CSE 107) · Classical & Bayesian Inference (STAT 132) · Applied Discrete Mathematics (CSE 16) · Computer Systems & Assembly (CSE 12) · Financial Mathematics (MATH 114) · Introduction to R (STAT 266A)